

PHYS 4003: Advanced Mechanics

Michael Cai

March 30, 2025

Abstract

Goldstein (2nd Edition)

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Chapter 1

Course Overview

Lecture 1

1.1 Topics to be Covered

- Brief review of Newtonian mechanics
- Discuss constraints and the first formulation of Lagrangian mechanics
- Reformulate in terms of Hamilton's principle (principle of least action)
- Explore invariants of the action and Lagrangian, and their connection to conserved quantities
- Explore applications
- Reformulate Classical Mechanics using the Hamiltonian
- Canonical transformations (Hamilton-Jacobi Theory)

1. Point Particles

Position and Velocity: Denote the position of a particle by $\vec{r}$ and its velocity by

$$\vec{v} = \dot{\vec{r}} = \frac{d\vec{r}}{dt}.$$

For point particles, we assume they are sizeless and that there is no intrinsic angular momentum.

Momentum: Define the momentum as

$$\vec{p} = m\vec{v},$$

with the assumption that $\dot{m} = 0$ (mass is constant).

Newton's Second Law: Newton's second law is given by

$$\vec{F} = \dot{\vec{p}} = m\vec{a} = m\ddot{\vec{r}} = m\frac{d^2\vec{r}}{dt^2},$$

where $\vec{F}$ is the total force acting on the particle. If $\vec{F} = 0$, then

$$\dot{\vec{p}} = 0,$$

which implies that both $\vec{p}$ and $\vec{v}$ are constant.

Work and Kinetic Energy: Define the work done from state 1 to state 2 as

$$W_{12} = \int_1^2 \vec{F} \cdot d\vec{s}.$$

If we express $d\vec{s}$ in terms of time, we have

$$W_{12} = \int_1^2 dt \vec{v} \cdot m\dot{\vec{v}}.$$

Recognizing that

$$\vec{v} \cdot \dot{\vec{v}} = \frac{1}{2} \frac{d}{dt} (\vec{v}^2),$$

we can write

$$W_{12} = \frac{m}{2} \int_1^2 dt \frac{d}{dt} (\vec{v}^2) = T_2 - T_1,$$

where the kinetic energy is defined as

$$T = \frac{1}{2} m \vec{v}^2.$$

Conservative Forces and Energy Conservation: If the force $\vec{F}$ is conservative, then there exists a scalar potential $V(\vec{r})$ such that

$$\vec{F} = -\nabla V(\vec{r}).$$

In this case,

$$\vec{F} \cdot d\vec{s} = -\nabla V \cdot d\vec{s},$$

and conservation of energy leads to

$$T_1 + V_1 = T_2 + V_2.$$

Metric and Coordinate Systems: Notice that the quantity $\vec{r} \cdot \vec{r}$ is invariant under rotations. This allows us to specify coordinates using a chosen metric. For example, in Cartesian coordinates we have

$$ds^2 = dx^2 + dy^2 + dz^2,$$

while in spherical coordinates the line element is given by

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

Chapter 2

Galilean Transformations and Systems of Particles

Lecture 2:

In these notes we explore the invariance of Newton's laws under Galilean transformations, derive the admissible coordinate transformations, and then extend our discussion to systems of particles and constraints.

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2.1 Galilean Transformations

Consider a (Galilean) transformation of the form

$$\mathbf{r} = \mathbf{r}' + \Delta,$$

or, equivalently,

$$\mathbf{r}' = \mathbf{r} - \Delta(t).$$

Differentiating with respect to time, the velocities transform as

$$\mathbf{v}' = \mathbf{v} - \dot{\Delta},$$

and the accelerations as

$$\mathbf{a}' = \mathbf{a} - \ddot{\Delta}.$$

We demand that the second derivative of the displacement vanishes, i.e.,

$$\ddot{\Delta} = 0,$$

so that

$$\Delta(t) = \mathbf{v}_G t,$$

where $\mathbf{v}_G$ is called the *Galilean velocity*.

Next, suppose we have Cartesian coordinates $x = \{x_1, x_2, x_3\}$ and we introduce a new coordinate system given by

$$x'_i = f_i(x, t).$$

Taking the time derivative, we obtain

$$\dot{x}'_i = \sum_j \frac{\partial f_i}{\partial x_j} \dot{x}_j + \frac{\partial f_i}{\partial t}.$$

Differentiating once more to obtain the acceleration, we have

$$\ddot{x}'_i = \frac{d}{dt} \left(\sum_j \frac{\partial f_i}{\partial x_j} \dot{x}_j \right) + \sum_j \frac{\partial^2 f_i}{\partial x_j \partial t} \dot{x}_j + \frac{\partial^2 f_i}{\partial t^2}.$$

Expanding the total time derivative gives

$$\frac{d}{dt} \left(\sum_j \frac{\partial f_i}{\partial x_j} \dot{x}_j \right) = \sum_j \frac{\partial f_i}{\partial x_j} \ddot{x}_j + \sum_j \sum_k \frac{\partial^2 f_i}{\partial x_j \partial x_k} \dot{x}_j \dot{x}_k.$$

Thus, the full expression becomes

$$\ddot{x}'_i = \sum_j \frac{\partial f_i}{\partial x_j} \ddot{x}_j + 2 \sum_j \frac{\partial^2 f_i}{\partial x_j \partial t} \dot{x}_j + \sum_j \sum_k \frac{\partial^2 f_i}{\partial x_j \partial x_k} \dot{x}_j \dot{x}_k + \frac{\partial^2 f_i}{\partial t^2}.$$

Eliminating Acceleration in the New Frame: Since we assume that there are no forces in the original frame (i.e., $\ddot{x}_j = 0$), the first term vanishes. To ensure that the transformed acceleration $\ddot{x}'_i$ vanishes for all motions, we require that each of the remaining terms vanish separately:

1. The third term involves $\frac{\partial^2 f_i}{\partial x_j \partial x_k}$. To make this vanish, we must have

$$\frac{\partial^2 f_i}{\partial x_j \partial x_k} = 0,$$

so that f_i is *linear* in the coordinates. Thus, we can write

$$f_i(x, t) = \sum_k a_{ik}(t) x_k + \beta_i(t).$$

2. Next, to eliminate the second term, notice that it involves the time derivative of the coefficients $a_{ik}(t)$. Therefore, we require

$$\frac{\partial a_{ik}(t)}{\partial t} = 0,$$

so that the matrix a_{ik} is time-independent.

3. Lastly, to eliminate the fourth term, we require

$$\frac{\partial^2 \beta_i(t)}{\partial t^2} = 0,$$

which implies that $\beta_i(t)$ is linear in time.

In summary, the admissible coordinate transformations under the Galilean symmetry are of the form

$$x'_i = \sum_k a_{ik} x_k + v_{G,i} t,$$

where the constants a_{ik} represent rotations (or, more generally, linear transformations preserving the structure of space) and $v_{G,i}$ are the components of the Galilean velocity.

2.2 Systems of Particles

Now consider a system of N particles in a Cartesian configuration space. Let the positions be $\{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N\}$. The forces acting on particle i are given by Newton's law:

$$\mathbf{F}_i = m_i \mathbf{a}_i = \dot{\mathbf{p}}_i,$$

and may be decomposed as

$$\mathbf{F}_i = \mathbf{F}_i^{(e)} + \mathbf{F}_i^{(i)},$$

where $\mathbf{F}_i^{(e)}$ are external forces and $\mathbf{F}_i^{(i)}$ are internal forces. Summing over all particles, we have

$$\dot{\mathbf{p}} = \sum_i \mathbf{F}_i^{(e)} = \mathbf{F}^{(e)},$$

and by Newton's third law the internal forces satisfy

$$\sum_i \mathbf{F}_i^{(i)} = 0.$$

Kinetic Energy and Work: The total kinetic energy is given by

$$T = \sum_i \frac{1}{2} m_i \mathbf{v}_i^2.$$

Taking the time derivative yields

$$\frac{dT}{dt} = \sum_i m_i \mathbf{v}_i \cdot \dot{\mathbf{v}}_i = \sum_i \mathbf{v}_i \cdot \mathbf{F}_i,$$

which represents the power delivered to the system. Integrating over time from t_1 to t_2 we obtain

$$T_2 - T_1 = \int_{t_1}^{t_2} \frac{dT}{dt} dt = \int_{t_1}^{t_2} dt \sum_i \mathbf{v}_i \cdot \mathbf{F}_i.$$

Since the work done is defined by

$$W_{12} = \int_1^2 \mathbf{F} \cdot d\mathbf{s},$$

this shows that the change in kinetic energy is equal to the work done on the system:

$$T_2 - T_1 = W_{12}.$$

If the force on each particle is conservative, meaning

$$\mathbf{F}_i = -\nabla_i V(\mathbf{r}, t),$$

then the total mechanical energy (kinetic plus potential) is conserved.

2.3 Constraints

Each constraint in a mechanical system removes one degree of freedom. A *holonomic constraint* can be written as

$$f(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t) = 0.$$

If there are k such holonomic constraints, then the number of independent degrees of freedom is reduced from $3N$ to $3N - k$. We introduce a set of generalized coordinates $\{q_1, q_2, \dots, q_{3N-k}\}$ so that the positions become functions of these coordinates and time:

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \dots, q_{3N-k}, t).$$

Then the velocities are given by

$$\mathbf{v}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t}.$$

Notice that

$$\frac{\partial \mathbf{v}_i}{\partial \dot{q}_j} = \frac{\partial \mathbf{r}_i}{\partial q_j}.$$

It is also convenient to define the generalized force corresponding to the coordinate q_i as

$$Q_i = \sum_j \mathbf{F}_j \cdot \frac{\partial \mathbf{r}_j}{\partial q_i}.$$

If the forces are derivable from a potential $V(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t)$, then by the chain rule the derivative of the potential with respect to the generalized coordinate is

$$\frac{\partial V}{\partial q_i} = \sum_j \nabla_j V \cdot \frac{\partial \mathbf{r}_j}{\partial q_i}.$$

Summary of Explanations:

- **Galilean Transformations:** We demonstrated that the most general coordinate transformations preserving the form of Newton's laws (i.e., leaving the acceleration invariant) are linear in the spatial coordinates and involve a translation proportional to time. These transformations form the basis of Galilean invariance.
- **Systems of Particles:** By considering the total kinetic energy and its time derivative, we showed that the work done on the system is equal to the change in kinetic energy. When forces are conservative, the total energy is conserved.
- **Constraints:** Constraints reduce the number of independent coordinates in a system. By introducing generalized coordinates, the equations of motion can be reformulated in terms of these independent variables, and the generalized forces are defined accordingly.

Chapter 3

Lagrangian Mechanics

Lecture 3

3.1 Derivation of Euler Lagrange Equations

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Remark. As a recap of last lecture we have N particles which lead to degrees of freedom of $3N$ with K constraints leading to generalized coordinates of $3N - k$. For any system, there are an infinite number of generalized coordinates.

Recall that we can write the movement of these particles more simply

$$\vec{r}_i(q_1, q_2, \dots, q_{3N-k}, t)$$

This helps reestablish Newton's laws in generalized coordinates that can have curvature. We see this through the derivative with respect to a generalized coordinate $\frac{\partial \vec{r}}{\partial q_j}$

We thus have to restate the motion of the particle in the said new coordinates. Recall that the velocity of the i th particle would be

$$\vec{v}_i = \sum_j \frac{\partial \vec{r}_j}{\partial q_j} \frac{dq_j}{dt} + \frac{\partial \vec{r}_i}{\partial t}$$

From here, we can apply this to a 2D problem which respects the constraint that $z = 0$. Specify the angle and distance as θ, r with the orthogonal unit vectors. In our generalized coordinates we can write out

$$x = r \cos \theta, y = r \sin \theta$$

We can write the full position vector as

$$\vec{r} = x\hat{x} + y\hat{y}$$

$$\frac{\partial \vec{r}}{\partial r} = \cos \theta \hat{x} + \sin \theta \hat{y}$$

$$\frac{\partial \vec{r}}{\partial \theta} = -r \sin \theta \hat{x} + r \cos \theta \hat{y}$$

We see this involves the generalized coordinates and scales by the length of the particle from the origin and so on. These derivatives are asking how these position vectors are differentially changing. These changes are vectors that tell how the system changes as we change r and θ . One issue is that they don't necessarily have to have lengths of 1. Therefore, we can simply divide by its magnitude to find the unit vectors. Going back to the velocity vector we have

$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

We have that

$$Q_i = \sum_j \vec{F}_j \cdot \frac{\partial \vec{r}_j}{\partial q_i}$$

This is saying that given an infinitesimal change, what does it do to all the particles in the system i.e. changing to position vector. This is the force associated with the generalized coordinates. We now want to look at deriving the Work Energy Theorem

Remark. A trick to avoid the time evolution of the Work Energy theorem is to introduce virtual displacements on each of the particles. We label these as $\delta \vec{r}_i$. There are a whole set that are arbitrary but allow us to do experiments on a physical system. We can then calculate the virtual work under those displacements. We define the following:

$$\delta W = \sum_i \vec{F}_i \cdot \delta \vec{r}_i$$

These virtual displacements lead to virtual work in the generalized coordinates in the following way:

$$\delta \vec{r}_i = \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \cdot \delta q_j, \sum_j Q_j \delta q_j = \delta W$$

Through the geometrical factor, we can connect the euclidean vectors describing the system to what happens with the new generalized coordinates. We can then separate the forces into two kinds: applied and internal. We wish to remove the internal forces from our analysis where

$$\vec{F}_i = \vec{F}_i^a + \vec{F}_i^c$$

We can write

$$\delta W = \sum_i \vec{F}_i^a \cdot \delta \vec{r}_i + \sum_i \vec{F}_i^c \cdot \delta \vec{r}_i$$

We see that the constrain force is orthogonal to the direction of movement. One example is the normal force. Thus, any displacement we imagine will have a dot product equal to 0. One counter example is static friction force is, the dot product could be nonzero.

We can thus choose to write that the second term is zero $\delta W^c = 0$. Note that constraint forces are not defined by a potential energy function. It must depend on other things other than the position of the object $\vec{r}$. We can now ignore all forces of constraint!

$$\vec{F}_i = \vec{p}_i$$

$$\sum_i (\vec{F}_i - \vec{p}_i) \cdot \delta \vec{r}_i = 0$$

This is simple rearrangement and nothing has been done. One has to be careful here since the i's are correlated since the constraints determine the position vectors in a smaller set of generalized coordinates, we can make a statement for the sum. We have done a sleight of hand by neglecting notationally the constraint force. Thus, we only have the applied forces and the $\delta \vec{r}$ are correlated, they apply for the sum and not term by term. We will choose to manipulate the following:

$$\begin{aligned} \delta W &= \sum_i \vec{p}_i \cdot \delta \vec{r}_i \\ &= \sum_{i,j} \vec{p}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j \end{aligned}$$

which relates the cartesian coordinates to the generalized coordinates. We can manipulate this to see what it would look like for one of the i'th particles. We will have

$$\vec{p}_i \frac{\partial \vec{v}_i}{\partial q_j} = m_i \dot{v}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j}$$

where we see that $\frac{\partial \vec{r}_i}{\partial q_j}$ represents the geometry of the coordinates. Note that we must consider the change in the coordinates itself as shown in the polar case. Thus, we will have the simplification to be

$$\frac{d}{dt}(m_i \vec{v}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j}) - m_i \vec{v}_i \cdot \frac{d}{dt} \frac{\partial \vec{r}_i}{\partial q_j}$$

Continuing in simplifying the second term we have that

$$\begin{aligned} \frac{d}{dt} \frac{\partial \vec{r}_i}{\partial q_j} &= \sum_k \frac{\partial^2 \vec{r}}{\partial q_j \partial q_k} \dot{q}_k + \frac{\partial^2 \vec{r}}{\partial t \partial q_j} \\ &= \frac{\partial}{\partial q_j} \left[\sum_k \frac{\partial \vec{r}_i}{\partial q_k} \dot{q}_k + \frac{\partial \vec{r}_i}{\partial t} \right] \end{aligned}$$

where inside the braces represents the velocity of the i'th particle. Thus our result will be

$$= \frac{\partial \vec{v}_i}{\partial q_j}$$

Recall the relationship where

$$\frac{\partial \vec{r}_i}{\partial q_j} = \frac{\partial \vec{v}_i}{\partial \dot{q}_j}$$

Now use the fact that $T_i = \frac{1}{2} m_i \vec{v}_i^2$. Thus, we can get the result that

$$\frac{\partial T_i}{\partial q_j} = m \vec{v}_j \cdot \frac{\partial \vec{v}_i}{\partial q_j}$$

Going back we can rewrite our results as

$$\vec{p}_i \frac{\partial \vec{r}_j}{\partial q_j} = \frac{d}{dt} \left(\frac{\partial T_i}{\partial \dot{q}_j} \right) - \frac{\partial T_i}{\partial q_j}$$

Now applying this to our whole system we have that

$$\begin{aligned} 0 &= \sum_{i,j} \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j} \delta q_j - \left[\frac{d}{dt} \left(\frac{\partial T_i}{\partial \dot{q}_j} \right) - \frac{\partial T_i}{\partial q_j} \right] \delta q_j \\ 0 &= \sum_j \left[Q_j - \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) + \frac{\partial T}{\partial q_j} \right] \delta q_j \end{aligned}$$

If δq_j unconstrained, then the square brackets must be 0 term by term based on the virtual displacement. Thus, for j'th coordinates, we will have the relationship that

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j$$

This is the form for generalized coordinates. Now assume that there exists a defined potential with conservative forces s.t.

$$-\frac{\partial V}{\partial q_j} = Q_j$$

Assuming $V(q, t)$, we can define the Lagrangian $L = T - V$ to get the Euler Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0$$

Lecture 4

Recall

For a system with generalized coordinates q_j and kinetic energy T , the Euler–Lagrange equation is

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} = Q_j,$$

where the generalized force Q_j is given by

$$Q_j = \sum_i \vec{F}_i \cdot \frac{\partial \vec{r}_i}{\partial q_j}.$$

If the forces $\vec{F}_i$ are derivable from a potential $V(\vec{r}_1, \dots, \vec{r}_N)$, then

$$Q_j = -\frac{\partial V}{\partial q_j}.$$

In that case, the Euler–Lagrange equation becomes

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = 0,$$

where the Lagrangian is $L = T - V$.

3.2 Polar Coordinates Example

In polar coordinates, we relate Cartesian coordinates (x, y) to polar coordinates (r, θ) by

$$x = r \cos \theta, \quad y = r \sin \theta.$$

The partial derivatives with respect to r are

$$\frac{\partial x}{\partial r} = \cos \theta, \quad \frac{\partial y}{\partial r} = \sin \theta,$$

and with respect to θ are

$$\frac{\partial x}{\partial \theta} = -r \sin \theta, \quad \frac{\partial y}{\partial \theta} = r \cos \theta.$$

Thus, the generalized forces in polar coordinates are:

$$Q_r = \vec{F} \cdot \frac{\partial \vec{r}}{\partial r} = F_x \cos \theta + F_y \sin \theta = \vec{F} \cdot \hat{r},$$

$$Q_\theta = \vec{F} \cdot \frac{\partial \vec{r}}{\partial \theta} = -F_x(r \sin \theta) + F_y(r \cos \theta) = r \vec{F} \cdot \hat{\theta}.$$

The kinetic energy in polar coordinates is given by

$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2).$$

Its derivatives are:

$$\begin{aligned} \frac{\partial T}{\partial \dot{r}} &= m\dot{r}, & \frac{\partial T}{\partial r} &= mr\dot{\theta}^2, \\ \frac{\partial T}{\partial \dot{\theta}} &= mr^2\dot{\theta}, & \frac{\partial T}{\partial \theta} &= 0. \end{aligned}$$

Thus, applying the Euler–Lagrange equations:

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{r}} - \frac{\partial T}{\partial r} = Q_r,$$

leads to

$$m\ddot{r} - mr\dot{\theta}^2 = \vec{F} \cdot \hat{r},$$

and for the θ coordinate,

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{\theta}} - \frac{\partial T}{\partial \theta} = Q_\theta,$$

which gives

$$mr^2\ddot{\theta} + 2mrr\dot{\theta} = r \vec{F} \cdot \hat{\theta}.$$

If we suppose that the potential V depends only on r , then the Euler–Lagrange equation for r becomes

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{r}} - \frac{\partial T}{\partial r} + \frac{\partial V}{\partial r} = 0.$$

3.3 Generalized Momentum and Cyclic Coordinates

The generalized momentum p_j is defined as

$$p_j = \frac{\partial L}{\partial \dot{q}_j},$$

which, for the kinetic energy, is usually

$$p_j = \frac{\partial T}{\partial \dot{q}_j}.$$

If the Lagrangian does not depend on a coordinate q (i.e., q is cyclic), then the corresponding momentum is conserved. For polar coordinates, since

$$\frac{\partial T}{\partial \dot{\theta}} = mr^2 \dot{\theta},$$

the angular momentum $p_\theta = mr^2 \dot{\theta}$ is conserved when θ is a cyclic coordinate.

3.4 Generalized Potentials

It is possible to have a potential that depends on the velocities $\dot{q}_j$ and still satisfy the Euler–Lagrange equations. Consider a generalized potential $U(q, \dot{q}, t)$ such that

$$Q_j = -\frac{\partial U}{\partial q_j} + \frac{d}{dt} \frac{\partial U}{\partial \dot{q}_j}.$$

Then, with the Lagrangian defined as

$$L = T - U,$$

we still have

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} = Q_j.$$

3.5 Electromagnetic Interactions

For electromagnetic (EM) interactions, the Lagrangian is given by

$$L = T - q\phi - \frac{q}{c} \vec{A} \cdot \vec{v},$$

where q is the charge, ϕ is the scalar potential, $\vec{A}$ is the vector potential, and c is the speed of light.

Using Gaussian units, Maxwell's equations are:

$$\begin{aligned} \nabla \cdot \vec{E} &= 4\pi\rho, \\ \nabla \times \vec{E} &= -\frac{1}{c} \frac{\partial \vec{B}}{\partial t}, \\ \nabla \cdot \vec{B} &= 0, \\ \nabla \times \vec{B} &= \frac{4\pi}{c} \vec{J} + \frac{1}{c} \frac{\partial \vec{E}}{\partial t}. \end{aligned}$$

The potentials are related to the fields by

$$\vec{B} = \nabla \times \vec{A}, \quad \vec{E} = -\nabla\phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t}.$$

The electromagnetic force is given by

$$\vec{F}_{em} = q\vec{E} + \frac{q}{c} \vec{v} \times \vec{B}.$$

Substituting the expressions for $\vec{E}$ and $\vec{B}$, we obtain

$$\vec{F}_{em} = q \left(-\nabla\phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} + \frac{1}{c} \vec{v} \times (\nabla \times \vec{A}) \right).$$

After carrying out the necessary algebra (i.e., expanding in components), one finds that

$$\vec{F}_{em} = q \left(-\nabla\phi - \frac{1}{c} \frac{d\vec{A}}{dt} \right).$$

This suggests a generalized potential of the form

$$U = q \left(\phi - \frac{1}{c} \vec{v} \cdot \vec{A} \right),$$

so that the Lagrangian becomes

$$L = T - U = T - q\phi + \frac{q}{c} \vec{v} \cdot \vec{A}.$$

Chapter 4

Variational Calculus

Lecture 5

4.1 Gauge Transformation

Jan 28 01:10

Remark. Why are the Euler Lagrange equations in a specific structure that makes the physics unchanged by adding a total time derivative. This suggests that something is going on other than the brute force derivation of curvilinear coordinates. It seems that there is something deeper going on here.

Remark. From last lecture the electromagnetic term we get is

$$L_{EM} = T - q\phi + \frac{q}{c} \vec{v} \cdot \vec{A}$$

Note if you factor out the q and a negative sign we get that

$$L_{EM} = T - q\left(\phi - \frac{1}{c} \vec{v} \cdot \vec{A}\right)$$

which looks like the lorentz product of a lorentz velocity and a lorentz potential. The problem though is that there are ambiguities in the potential which is in a gauge transformation.

$$g_T : \phi \rightarrow \phi' = \phi - \frac{1}{c} \frac{\partial \psi}{\partial t}, \psi(\vec{r}, t)$$

$$g_T : A \rightarrow A' = \vec{A} + \vec{\nabla} \psi$$

$$\vec{B}' = \vec{\nabla} \times \vec{A}' + \nabla \times (\nabla \psi) = \vec{B}$$

We can check this for the electric field where

$$\begin{aligned} \vec{E}' &= -\nabla \phi' + \frac{1}{c} \frac{\partial \vec{A}'}{\partial t} \\ &= -\nabla \phi + \frac{1}{c} \nabla \frac{\partial \psi}{\partial t} - \frac{1}{c} \left(\frac{\partial \vec{A}}{\partial t} + \frac{\partial}{\partial t} \nabla \psi \right) \end{aligned}$$

Thus we get the original electric field $\vec{E}$. Thus we get an ambiguity in the potentials and we can change the electric potential or electric potential and the physics is unmodified. Now we ask the question of what happens to the Lagrangian L_{EM} ?

$$L'_{EM} = T - q\phi' + \frac{q}{c} \vec{v} \cdot \vec{A}'$$

$$\begin{aligned}
&= T - q\phi + \frac{q}{c} \frac{\partial \psi}{\partial t} + \frac{q}{c} \vec{v} \cdot (\vec{A} + \nabla \psi) \\
&= L_{EM} + \frac{q}{c} \left(\frac{\partial \psi}{\partial t} + \vec{v} \cdot \nabla \psi \right)
\end{aligned}$$

We look back where

$$\frac{d\psi}{dt} = \frac{\partial \psi}{\partial t} + \frac{\partial \psi}{\partial x} \hat{x} \cdots = \vec{v} \cdot \nabla \psi$$

Therefore we have that

$$L'_{EM} = L_{EM} + \frac{q}{c} \frac{d\psi}{dt}$$

There is something going on where under the transformation where

$$\frac{d}{dx} \left[\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} \right] = 0$$

we recast our understanding of a knowledge of why everything works. We will now motivate what we see by seeing what the other way is to see why the structure has a understanding of nontrivial aspects of the physics.

4.2 Calculus of Variations Derivation of E-L

Remark. The Euler Lagrange falls out of the calculus of variations. We can say that the state of a system is specified by the configuration space of the generalized coordinates q . Now we start by specifying the state of the system at two times T_1 and T_2 where $q_1 \equiv q(t_1)$ and $q_2 \equiv q(t_2)$. Now we ask how does the system evolve under the time evolution?

We will reformulate mechanics to consider all paths in $q(t)$ and consider all possible paths in configuration space from $q_1(t_1) \rightarrow q_2(t_2)$. We then ask which is the physical path the particle takes? We start by defining the action which is defined to be

$$I = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$$

The bounding principle is going to be defined as follows

Definition 4.2.1. Hamilton's Principle: The physical path is that for which the action I is stationary or "minimum". Basically, as we vary the paths in a controlled way, what is the variation δI

$$\delta I = \delta \int_{t_1}^{t_2} L dt = 0$$

all paths in $q(t)$ close to the physical path will have approximately the same action.

The next step is to see that the setup of the Euler-Lagrange equation is very general. If you consider different paths from one point to another and define a function integral and you ask what is the path for which the integral is stationary you get the Euler Lagrange equations. What is special is simply being the equations of motion to be the lagrangian. Note that the variational principle is slightly different though.

Remark. The crucial step is to specify the beginning and end points. All variations in paths respect this idea. Let's imagine all possible paths

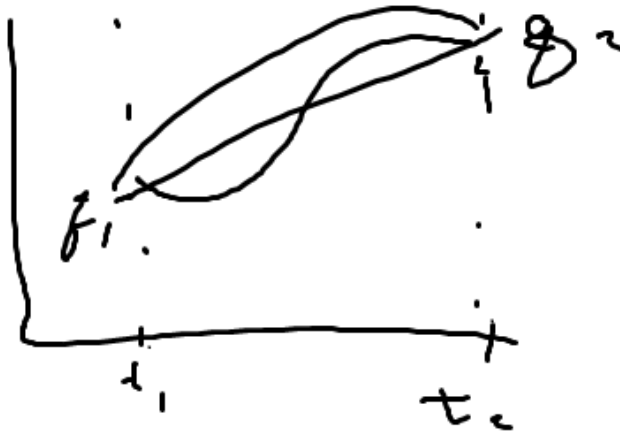


Figure 4.1

we only consider small changes from the physical path to determine if it is stationary. An important note is that the slope of the trajectory is the velocity which determines $q(t)$ and determines $\dot{q}$. Now consider an integral of an arbitrary function $f(q, \dot{q}, t)$. Where we define

$$J \equiv \int_{t_1}^{t_2} f(q, \dot{q}, t) dt$$

where it is self evident that it is a self generalization of the action. Now what conditions on the path is J stationary, i.e. $\delta J = 0$

Now we introduce $q(t, \alpha)$ where α is a parameter that describes which path to choose. We reiterate that

$$q(t_1, \alpha) = q_1, q(t_2, \alpha) = q_2, \forall \alpha$$

Therefore we get

$$J(\alpha) = \int_{t_1}^{t_2} f(q(t, \alpha), \dot{q}(t, \alpha), t) dt$$

Thus we specify that

$$\frac{dJ}{d\alpha} = 0$$

Now that α parameterized the paths. The derivative with respect to α should be 0 for the stationary states. Now this is so for when $\alpha = 0$ which gives a stationary choice. Since we are interested in small variation, we have

$$q(t, \alpha) \approx q(t, 0) + \left. \frac{\partial q}{\partial \alpha} \right|_{\alpha=0} \alpha + \mathcal{O}(\alpha^2)$$

Since these are the paths close to the stationary one, we can Taylor expand it. Let us call the derivative

$$\frac{\partial q}{\partial \alpha} = \eta(t)$$

note that any deviation from the path at t_1 and t_2 are going to be 0. Thus we notice that $\eta(t_1) = \eta(t_2) = 0$ We can rewrite

$$q = q(t, 0) + \eta\alpha$$

$$\dot{q} = \dot{q}(t, 0) + \dot{\eta}\alpha$$

From here we would like to calculate

$$\frac{dJ}{d\alpha} = \frac{d}{d\alpha} \int_{t_1}^{t_2} f(q, \dot{q}, t) dt$$

where Feynman made it popular to differentiate inside the integral. In this case we can rewrite this since α doesn't know about time

$$\begin{aligned} &= \int_{t_1}^{t_2} \frac{df}{d\alpha} dt \\ &= \int_{t_1}^{t_2} \left[\frac{\partial f}{\partial q} \frac{\partial q}{\partial \alpha} + \frac{\partial f}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial \alpha} \right] dt \end{aligned}$$

This form arises simply from the functional form. The functional dependence is all there is. We can rewrite this as

$$= \int_{t_1}^{t_2} \left[\frac{\partial f}{\partial q} \eta + \frac{\partial f}{\partial \dot{q}} \dot{\eta} \right] dt$$

this structure is just determined by the functional dependence on the coordinates. The next crucial step is to integrate the second term by parts by moving the time derivative.

$$= \int_{t_1}^{t_2} \left[\frac{\partial f}{\partial q} \eta - \frac{d}{dt} \left(\frac{\partial f}{\partial \dot{q}} \right) \eta \right] + \frac{\partial f}{\partial \dot{q}} \eta \Big|_{t_1}^{t_2} dx$$

however recall that the boundary terms of η are 0 and therefore the second term must be 0. Now under what conditions must we have that it is 0.

$$\frac{dJ}{d\alpha} = \int_{t_1}^{t_2} \left[\frac{\partial f}{\partial q} - \frac{d}{dt} \left(\frac{\partial f}{\partial \dot{q}} \right) \right] \eta dt = 0$$

notice we never said what η is except it is a function of time. Note that this must be 0 for any function we put in here. Basically,

$$\frac{dJ}{d\alpha} = 0 \text{ for all possible } \eta(t)$$

Now this gives the fundamental lemma of variational calculus. Thus, the quantity in the square brackets must be 0. We conclude that

$$\frac{\partial f}{\partial q} - \frac{d}{dt} \left[\frac{\partial f}{\partial \dot{q}} \right] = 0$$

This is not considering a physical system but rather a coordinate at two times and all possible paths as well as an arbitrary function that depends on 3 terms. Integrated it, and got completely general results with the above form. Thus everything comes down to the coordinate dependence and the shape of the path determines the relationship between q and $\dot{q}$. Now we think that the lagrangian depends on coordinates, velocity, and time. Now imagine if we had a function dependent on a second time derivative, we would have an additional term. If we connect this back to physics, the reason why we have the Euler Lagrange form is because it has no dependence on the second order of the coordinates. This structure is thus intimately connected with the coordinate dependence of the lagrangian and we can reformulate classical mechanics in terms of the Hamiltonian where th

Lecture 6

4.3 Claculus of Variations

Jan 28 01:10

Recall that

$$J = \int_{t_1}^{t_2} f(q, \dot{q}, t) dt$$

$$\delta J = 0 \rightarrow \text{Stationary}$$

The path is determined by the solution to

$$\frac{d}{dt} \left(\frac{\partial f}{\partial \dot{q}} \right) - \frac{\partial f}{\partial q} = 0$$

We would like to look at the situation where the term

$$\frac{\partial f}{\partial q} = 0$$

If f doesn't depend on q , then we will get the statement that

$$\frac{d}{dt} \left(\frac{\partial f}{\partial \dot{q}} \right) = 0$$

Here this isn't exactly classical mechanics, this is a trivial integration where we see that $\frac{\partial f}{\partial \dot{q}} = 0$. Note that there is something more interesting going on. Suppose both terms in the equation are non-zero and f doesn't depend on t where

$$f(q, \dot{q})$$

then there exists another conserved quantity and is written as follows:

$$h = \dot{q} \frac{\partial f}{\partial \dot{q}} - f$$

and if there is no time dependence then the quantity is constant.

$$\frac{dh}{dt} = \dot{q} \frac{\partial f}{\partial \dot{q}} + \dot{q} \frac{d}{dt} \left(\frac{\partial f}{\partial \dot{q}} \right)$$

This will then look like

$$\begin{aligned} &= -\frac{\partial f}{\partial q} \dot{q} - \frac{\partial f}{\partial \dot{q}} q'' \\ &= \dot{q} \left[\frac{d}{dt} \frac{\partial f}{\partial \dot{q}} - \frac{\partial f}{\partial q} \right] = 0 \end{aligned}$$

What we can do is say that h is a constant and already solved the problem

Example. Consider the x-y plane and the following figure

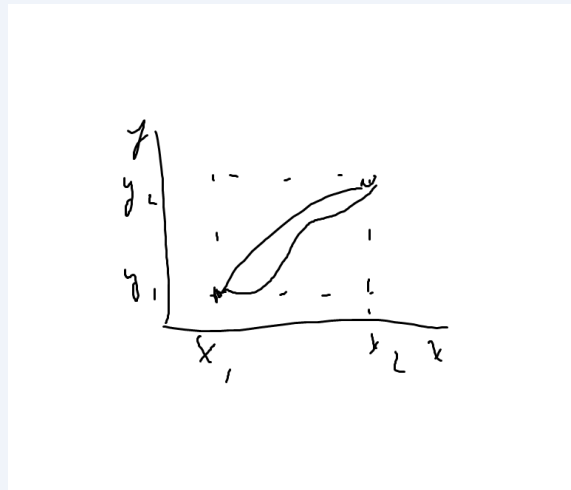


Figure 4.2

For this example it is quite easy to find the answer but we can't have the maximum path which is the extremum. However, we can carry out the expansion out to an order of α^2 if we want to check. The key thing to note is that the endpoints are fixed for this example.

$$\begin{aligned} J &= \int_{x_1}^{x_2} \sqrt{1 + (y')^2} dx \\ \delta J &= 0 \end{aligned}$$

If there are no explicit function on y then we immediately have that

$$\frac{d}{dt} \frac{\partial f}{\partial \dot{q}} = 0$$

We should have that

$$\frac{\partial f}{\partial \dot{y}} = C$$

$$\frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{1 + (y')^2}} = C$$

Often in the calculus of variations, the details matter. Notice that the quantity can never be larger than 1 and could be negative. Thus, the range of the function would be $(-1, 1)$

$$y^2(1 - C^2) = 1$$

$$y' = \frac{C}{\sqrt{1 - C^2}}$$

Let $a = \sqrt{1 - C^2}$. And we will have that

$$y = ax + b$$

Example. Brachistochrone: The calculus of variations were made here. We want the minimum time it takes to reach the point x

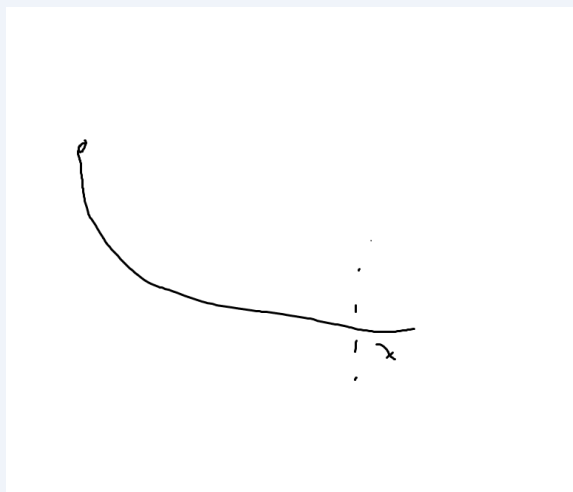


Figure 4.3

This isn't such a easy problem to understand the tradeoff between speed and distance moving. Thus, we wish to find the path for which t is the smallest. We can use a generalized coordinate where y is downwards.

$$\frac{1}{2} m \dot{y}^2 = mgy$$

$$\dot{y} = \sqrt{2gy}$$

We can write the functional integral to be

$$J = \int_{x_1}^{x_2} dt = \int_{x_1}^{x_2} \frac{s}{v} = \int_{x_1}^{x_2} \frac{\sqrt{1 + (y')^2}}{\sqrt{2gy}} dx$$

Therefore we have

$$f(y, y', x) = \sqrt{\frac{1 + (y')^2}{y}}$$

Note that there is an explicit dependence on y but not on x . Therefore we can write

$$h = y' \frac{1}{\sqrt{y}} \frac{y'}{\sqrt{1 + (y')^2}} - f = \text{Constant} = \frac{1}{C}$$

$$\begin{aligned} h &= \frac{1}{\sqrt{y}} \frac{1}{\sqrt{1 + (y')^2}} [(y')^2 - 1 - (y')^2] = \frac{1}{C} \\ &= -\frac{1}{\sqrt{1 + (y')^2}} \frac{1}{\sqrt{y}} = \frac{1}{C} \end{aligned}$$

$$y(1 + (y')^2) = C^2 \Rightarrow y' = \sqrt{\frac{C^2 - y}{y}}$$

This is more involved with a slope in terms of y and we can do some algebra to get a solution that the path that minimizes the time is a cycloid.

Now we can do this for a generalized coordinate in classical mechanics. We define the action as

$$I = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$$

which represents the full set of all q and $\dot{q}$ to avoid notation.

Definition 4.3.1. Hamilton's Principle:

The physical trajectory of the entire system for forever many generalized coordinates is where

$$\delta J = 0$$

We can parameterize the paths with α . We can parameterize the paths

$$q_j(t, \alpha) \rightarrow q_j(t, 0)$$

is going to be the physical path. We can expand these in a Taylor series:

$$q_j(t, \alpha) = q_j(t, 0) + \alpha \frac{\partial q_j}{\partial \alpha} \Big|_{\alpha=0}$$

This is a different function for each of the generalized coordinates where

$$\alpha \frac{\partial q_j}{\partial \alpha} = \eta_j$$

Note that

$$\eta_j(t_1) = \eta_j(t_2) = 0$$

We can take the derivative s.t.

$$\frac{dI}{d\alpha} = \int_{t_1}^{t_2} \frac{dL}{d\alpha} dt$$

The general terms of the calculus of variations is totally determined by the functional form of the position and the derivative. Nothing changes except that we have a sum of the coordinates and the derivatives.

$$= \int_{t_1}^{t_2} \sum_j \left[\frac{\partial L}{\partial q_j} \eta_j + \frac{\partial L}{\partial \dot{q}_j} \dot{\eta}_j \right] dt$$

Both terms are multiplied by η with a contact term that goes to 0. Thus this is going to be

$$= \int_{t_1}^{t_2} \sum_j \eta \left[\frac{\partial L}{\partial q_j} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} \right] dt + \sum_j \frac{\partial L}{\partial \dot{q}_j} \Big|_{t_1}^{t_2}$$

Note that the endpoints are 0 so therefore the last term is 0. Note that if we are holonomic and we respect the constraints, then the generalized coordinates are completely independent of each other. Now the reason why we use holonomic constraints because of the last crucial step. If we have nono-holonomic constraints or the coordinates don't satisfy the constraints, then the η_j terms are not independent and we cannot set the inside to be 0. Thus, the variations in the path are independent.

Now coming back recall

$$\frac{\partial L}{\partial q_j} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} = 0$$

Because

$$\delta I = 0$$

This screams that there is something simpler governing our physics. We then treat Hamilton's principle as the foundation of mechanics.

4.4 Generalized Momentum

Now suppose we define the generalized momentum to be

$$p_j \equiv \frac{\partial L}{\partial \dot{q}_j}$$

$$\frac{\partial L}{\partial q_j} = 0 \Rightarrow p_j = C$$

On a side note, suppose we have an isotropic universe with a universe that is the same everywhere. what would happen?

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$$

Suppose there is no external effect or forces. There are no potential that depends on x since the particle is isotropic.

$$p_x = m\dot{x}, p_y = m\dot{y}$$

Now Landau argues that in an inertial frame, we will have constant velocity and constant motion of a particle (homogeneous and isotropic universe) Now we have multiple coordinates and multiple terms for h to be

$$h = \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L$$

if $\frac{\partial L}{\partial t} = 0$ and therefore h is constant.

4.5 Respecting Constraints

What happens when the particles don't respect the constraints we want them to. What happens when the constrain force goes to 0? Now supposes that the q s don't respect our constraints or maybe constraint(s) satisfy the following:

$$f(q_1, \dots, q_N, \dot{q}_1, \dots, \dot{q}_N, t) = 0$$

$$\frac{dI}{d\alpha} = \int_{t_1}^{t_2} \sum_j \eta_j \left[\frac{\partial K}{\partial q_j} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} \right] dt$$

Suppose we do a minimization procedure such taht the constraints are forced to be satisfied as part of the solutions . We can introduce lagrangue multipliers that allow us to solve. We are going to introduce the lagrange multiplier into the action. Thus we use a modified action (principle) to enforce a constraint using λ_l .

$$I_C = \int_{t_1}^{t_2} \left(L - \sum_l \lambda_l f_l \right) dt$$

now we have a problem with more coordinates for the problem. In order to find the path that satisfies $\delta I = 0$. We now have to account for the λ . Now we can write the Euler Lagrange Equation as $\hat{L}$, thus we have

$$\frac{d}{dt} \left(\frac{\partial \hat{L}}{\partial \dot{\lambda}_l} \right) - \frac{\partial \hat{L}}{\partial \lambda_l} = 0$$

the first term is 0 because there is no derivative of λ and we thus know that

$$f_l = 0$$

which is exactly what we want since it must be enforced. Thus substituting in the lagrange multipliers works!

Lecture 7

4.6 Lagrange Multipliers

Feb 11 01:10

At the end of last lecture we started an analysis of systems where either the q s are not chosen to satisfy constraints automatically. Now suppose we had n generalized coordinates for a physical system and k constraints of the form

$$f_l((q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)) = 0$$

Then we included these in our variation procedure using Lagrange multipliers λ_l

$$I_c = \int_{t_1}^{t_2} \left[L + \sum_{l=1}^k \lambda_l(t) f_l \right] dt$$

where we call the term inside the brackets $\hat{L}$. But now we have to include the λ s as part of the variations but we already know their E-L equations to be

$$\frac{d}{dt} \left(\frac{\partial \hat{L}}{\partial \dot{\lambda}} \right) - \frac{\partial \hat{L}}{\partial \lambda} = 0$$

The first term in the E-L equation is zero and the second is just f_l so the E-L yields

$$f_l = 0$$

Now we have the E-L lagrange equations for $\hat{L}$. The lagrange multiplier technique enforces the constraint as part of the solution. But now for the q s we have additional terms of motion where it will be

$$-\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} + \frac{\partial L}{\partial q_j} + \sum_l \left[\lambda_l \frac{\partial f}{\partial q_j} - \frac{d}{dt} \left(\lambda_l \frac{\partial f}{\partial \dot{q}_j} \right) \right] = 0$$

where the summation is the generalized constraint force Q_j^c . Therefore we obtain the equation

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = Q_j^c$$

Remark. Even the non-holonomic constraints must do no virtual work.

Example. Rolling without slipping is a non-holonomic constraint example.

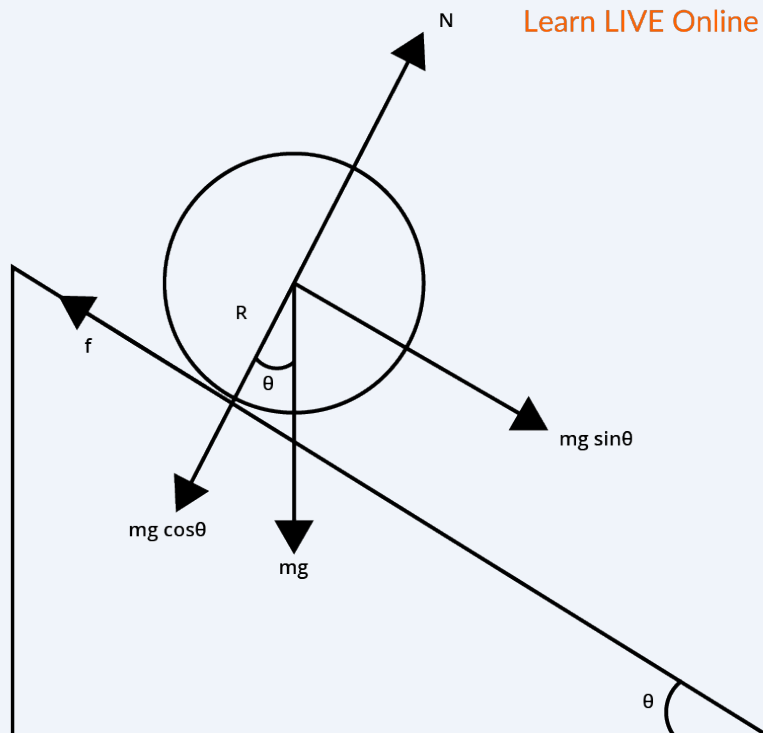


Figure 4.4

We can write that

$$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}mR^2\dot{\phi}^2$$

$$U = -mgx \sin \theta$$

$$L = T - U$$

The rolling without slipping constraint is denoted as

$$Rd\phi = dx$$

$$f = R\dot{\phi} - \dot{x} = 0$$

This is our non-holonomic constraint. So our generalized forces:

$$Q_x^c = \lambda \frac{\partial f}{\partial x} - \frac{d}{dt} \left(\lambda \frac{\partial f}{\partial \dot{x}} \right) = \dot{\lambda}$$

Similarly we have

$$Q_\phi^c = \lambda \frac{\partial f}{\partial \phi} - \frac{d}{dt} \left(\lambda \frac{\partial f}{\partial \dot{\phi}} \right) = -\dot{\lambda}R$$

Therefore we obtain the Euler lagrange equations to be

$$\dot{\lambda} = mx'' - mg \sin \theta$$

$$-R\dot{\lambda} = mR^2\phi''$$

to which we will add the constrain equation

$$R\dot{\phi} = \dot{x} \Rightarrow R\phi'' = x''$$

Substituting this in we have

$$\begin{aligned}\dot{\lambda} &= -x''m \\ -mx'' &= mx'' - mg \sin \theta \\ x'' &= \frac{1}{2}g \sin \theta\end{aligned}$$

And this is positive based on the coordinates we defined

$$\dot{\lambda} = -\frac{1}{2}mg \sin \theta = Q_x$$

This is a way to solve for non-holonomic constraint. However we showed it as a semi holonomic constraint. Note that the above differs in Goldstein in the sign of λ that $\dot{\lambda}$ appears instead of λ

Remark. A point of notation and physics: In deriving the E-L equaitons from Hamilton's principle, we expressed I in terms of variation parameter α then demanded that $\frac{dI}{d\alpha} = 0$ but suppose we define

$$\delta I = \frac{dI}{d\alpha} \delta \alpha$$

then

$$\begin{aligned}\delta I &= \int_{t_1}^{t_2} \sum_j \left[\frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \right] \eta_j \delta \alpha \, dt \\ \eta_j \alpha &= \delta q_j\end{aligned}$$

where we recall that

$$q_j(t, \alpha) \approx q_j(t, 0) + \eta_j \alpha$$

So we can rewrite Hamilton's principle

$$\delta I = \int_{t_1}^{t_2} \sum_j \left[\frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \right] \delta q_j \, dt$$

where δq_j is the same as variations when developing virtual work

Now why is this interesting? We have that

$$\begin{aligned}\delta I &= \delta \int_{t_1}^{t_2} (T - U) \, dt \\ &\Rightarrow \delta \int_{t_1}^{t_2} T \, dt - \delta \int_{t_1}^{t_2} U \, dt \\ &= \int_{t_1}^{t_2} \sum_j \left[\frac{\partial U}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial U}{\partial \dot{q}_j} \right) \right] \delta q_j \, dt\end{aligned}$$

where we observe how the generalized potentials fit in naturally. The inside of the brackets is the generalized force denoted as Q_j with a minus sign. Therefore, we can rewrite Hamilton's principle as

$$\delta \int_{t_1}^{t_2} T \, dt = - \int_{t_1}^{t_2} \sum_j Q_j \delta q_j \, dt$$

where the left hand is the different in integral of T w.r.t. infinitesimal variations in path and the right hand side is the time integral of virtual work associated with variations in path.

4.7 Symmetries and Conservation Laws

We have already introduced the idea of cyclic coordinates for which

$$\frac{\partial L}{\partial q} = 0$$

then from the euler lagrange equations momentum is conserved. For cyclic coordiantes, We have also seen a quantity h_i

$$h \equiv \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L = \sum_j \dot{q}_j p_j - L$$

such that $\frac{dh}{dt} = 0 \Leftrightarrow \frac{\partial L}{\partial t} = 0$. Indeed, in our homework we will show that if L is cyclic in time, h is conserved. Are there other conserved quantities? How to find or show that they exist?

We will see multiple times that infinitesimal transformations play a crucial role in mechanics for identifying Symmetries. Consider the behavior of the Lagrangian under the set of infinitesimal transformations

$$q_i \rightarrow q'_i = q_i - \epsilon \eta_i(q, t)$$

where η is the full set of q s. We can invert this and substitute into L

$$L \rightarrow L' = L(q(q'), \dot{q}(q'), \dot{q}'))$$

where L' depends on ϵ through the two values stated. So in fact we have

$$L' \rightarrow L \Leftrightarrow \epsilon = 0$$

This means the original Lagrangian expressed in terms of $q', \dot{q}'$. In the following analysis, we will consistently keep terms to first power in ϵ . Starting with Taylor expansion of L

$$L'(\epsilon) = L'(0) + \left. \frac{dL'}{d\epsilon} \right|_{\epsilon=0} \epsilon + \mathcal{O}(\epsilon^2)$$

Now we have that

$$\frac{\partial L'}{\partial \epsilon} = \sum_j \left[\frac{\partial L'}{\partial q_j} \frac{\partial q_j}{\partial \epsilon} + \frac{\partial L'}{\partial \dot{q}_j} \frac{\partial \dot{q}_j}{\partial \epsilon} \right]$$

But when we evaluate at $\epsilon = 0, L' \rightarrow L$. Therefore we have taht

$$L' = L + \sum_j \left[\frac{\partial L'}{\partial q_j} \frac{\partial q_j}{\partial \epsilon} + \frac{\partial L'}{\partial \dot{q}_j} \frac{\partial \dot{q}_j}{\partial \epsilon} \right] \epsilon$$

Now q_j depends on ϵ through the inverse relationship

$$q_j = q'_j - \epsilon \eta_j(q, t)$$

however we need this expression in terms of $q_j(q', t, \epsilon)$ so we evaluate $\frac{\partial q_j}{\partial \epsilon}$, and $\frac{\partial \dot{q}_j}{\partial \epsilon} \dots$

$$\frac{\partial q_j}{\partial \epsilon} = -\eta_j, \frac{\partial \dot{q}_j}{\partial \epsilon} = -\dot{\eta}_j = -\frac{d}{dt} \left(\frac{\partial q_j}{\partial \eta} \right)$$

Now we can evaluate $\left. \frac{\partial L'}{\partial \epsilon} \right|_{\epsilon=0}$.

$$\left. \frac{\partial L'}{\partial \epsilon} \right|_{\epsilon=0} = \sum_j \left[\frac{\partial L}{\partial q_j} (-\eta_j) + \frac{\partial L}{\partial \dot{q}_j} (-\dot{\eta}_j) \right]$$

Simplifying some terms we will have that

$$L' = L - \frac{d}{dt} \left(\sum_j \frac{\partial L}{\partial \dot{q}_j} \eta_j \right) \epsilon$$

But suppose our transformation leaves the Lagrangian invariant : $L' = L$. Then the quantity $\sum_j \frac{\partial L}{\partial \dot{q}_j} \dot{q}_j$ is conserved. Now consider an N- particle system with only internal forces:

$$L = \sum_i \frac{1}{2} m_i \dot{\vec{r}}_i^2 - V(\vec{r}_{ij})$$

Now consider a translation in infinitesimal form in the direction $\hat{\omega}$:

$$x_j \rightarrow x'_j = x_j + \epsilon \hat{\omega}_x \Rightarrow x_i = x'_i - \epsilon \hat{\omega}_x$$

and so forth for the rest of the coordinates too. Clearly we have that

$$\dot{x}'_i = \dot{x}_i$$

and for the y and z coordinates too. Also observe that all $\vec{r}'_{ij} = \vec{r}_{ij}$. Therefore substituting the inverting expressions for x_i, y_i, z_i into the Lagrangian:

$$L' = \sum_i \frac{1}{2} m_i (\dot{x}'_i{}^2 + \dot{y}'_i{}^2 + \dot{z}'_i{}^2) + V(\vec{r}'_{ij})$$

Clearly $\frac{\partial L'}{\partial \epsilon} = 0$. Equivalently the transformation has left the form of L unchanged. Thus, there exists a conserved quantity called $\mathcal{Q}$

$$\mathcal{Q} = \sum_j \frac{\partial L}{\partial \dot{q}_j} \eta_j = \left(\sum_i m_i \vec{v}_i \right) \cdot \hat{\omega}$$

Therefore the invariance of L under translation implies conservation of momentum.

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Lecture 10

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Lecture 11

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4.8 Central Force Motion: Laplace Runge Lenz Vector

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$$\frac{d^2 U}{d\theta^2} + U = -\frac{m}{l^2 \mu^2} f(u)$$

$$f = -\frac{k}{r^2} = -k\mu^2$$

$$\frac{d^2 U}{d\theta^2} + U = \frac{mk}{l^2}$$

$$X \equiv U - \frac{mk}{l^2}$$

$$\frac{dX}{d\theta^2} = -X$$

$$X = A \cos(\theta - \theta_0)$$

$$\frac{1}{r} = \frac{mk}{l^2} + A \cos(\theta - \theta_0)$$

$$= \frac{mk}{l^2} (1 + e \cos(\theta - \theta_0))$$

$$e = \frac{Amk}{l^2}$$

Recall that $e > 1$ is a hyperbole, equal to 1 is parabolic, less than one is ellipse, and 0 is circular. We can see a different to get this:

$$\exists \vec{A} = \vec{p} \times \vec{L} - \frac{mk}{r} \vec{r}$$

This vector is conserved s.t.

$$\frac{d\vec{A}}{dt} = 0$$

$$\frac{d\vec{A}}{dt} = \dot{\vec{p}} \times \vec{L} + \vec{p} \times \dot{\vec{L}} - mk \frac{d\hat{r}}{dt}$$

Note that obviously that the second term is zero since angular momentum is conserved. Next consider that $F = -\frac{k}{r^2} \hat{r}$

$$\dot{\vec{p}} \times \vec{L} = -\frac{k}{r^2} \frac{\vec{r}}{r} \times (\vec{r} \times \vec{p})$$

Note that

$$a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$$

We can thus write this as

$$\dot{\vec{p}} \times \vec{L} = -\frac{mk}{r^2} ((\vec{r} \cdot \vec{r})\vec{r} - r^2 \dot{\vec{r}})$$

$$\vec{r} \cdot (\dot{\vec{r}} + r\dot{\theta}\hat{\theta}) = r\dot{r}$$

Therefore we can write this as

$$\dot{\vec{p}} \times \vec{L} = -\frac{mk}{r^3} [r\dot{r} - r^2\dot{\vec{r}}]$$

This simplifies into

$$\frac{mk}{r} r\dot{\theta}\hat{\theta}$$

We must have that

$$\frac{d\vec{A}}{dt} = mr(\dot{\theta}\hat{\theta} - \theta\dot{\theta}) = 0$$

$$\vec{A} \cdot \vec{L} = 0$$

which implies that $\vec{A}$ lies in the plane of the motion.

$$\vec{A} \cdot \vec{r} = rA \cos(\theta - \theta_0) = \vec{r} \cdot (\vec{p} \times \vec{L}) - mk$$

Note that $\vec{L} \cdot (\vec{r} \times \vec{p}) = l^2$

$$\begin{aligned} \frac{1}{r} &= \frac{mk}{l^2} + A \cos(\theta - \theta_0) \\ &= \frac{mk}{k^2} \left(1 + \frac{A}{mk} \cos(\theta - \theta_0) \right) \end{aligned}$$

When $\theta = \theta_0$ the second term is largest which implies $\frac{1}{r}$ is smallest. Since $\frac{A}{mk}$ is the eccentricity. What we learn is that A points to the angle where r is smallest which is the periapsis or perihelion and

$$A^2 = 2ml^2E + m^2k^2$$

Chapter 5

Rigid Body Motion

We start by discussing rotations as finite transformations and then infinitesimal form. A rotation is a point transformation and affects all vectors. Any vector $\vec{v}' = A\vec{v}$ where $\vec{v}$ is any vector. This is true for all transformations however very important for rotations.

There are two ways to view the rotation. One is passive where we rotate the axes where $x, y \rightarrow x', y'$ with a difference of θ . Note that $\vec{v}$ is invariant and thus angular momentum is also invariant. What we are simply doing is changing the components of the vector and not the actual vector itself. When we define an active rotation, we rotate the vector $\vec{v} \rightarrow \vec{v}'$ which changes the vector itself by an angle θ . Note that mathematically it is simply a difference in sign where

$$A^p(\theta) = A^a(-\theta)$$

Consider the unit vectors and there transformation $\hat{i}, \hat{k} \rightarrow \hat{i}', \hat{j}'$ where we have that

$$\begin{aligned}\hat{i}' &= (\hat{i}' \cdot \hat{i})\hat{i} + (\hat{i}' \cdot \hat{j})\hat{j} + (\hat{i}' \cdot \hat{k})\hat{k} \\ &= \alpha_1\hat{i} + \alpha_2\hat{k} + \alpha_3\hat{k}\end{aligned}$$

and such for $\hat{j}'$ and $\hat{k}'$. Note that these are not all independent and can be reduced down since we know in 2 dimensions we can simplify the terms down to $\cos\theta$ and $\sin\theta$ which is only one quantity specified which implies some constraints i.e. orthonormal unit vectors. We require an orthogonal basis

$$\begin{aligned}\hat{i}' \cdot \hat{j}' &= \hat{i}' \cdot \hat{k}' = \hat{k}' \cdot \hat{k}' = 0 \\ \hat{i}' \cdot \hat{i}' &= 1 \dots\end{aligned}$$

It makes intuitive since then we need three angles to define our rotation which are optimal. We have that

$$\vec{v} \cdot \hat{i}' = v_x\alpha_1 + v_y\alpha_2 + v_z\alpha_3 = v'_x$$

and such for $\vec{v} \cdot \hat{j}' = v'_y$. We have thus specified the relationship of $\vec{v}' = \vec{A}\vec{v}$. Thus we essentially understand what the operator A does. However, the notation is not very clean and thus we have the following notation:

$$\begin{aligned}(x, y, z) &\rightarrow (x_1, x_2, x_3) \\ \hat{i}, \hat{j}, \hat{k} &\rightarrow \hat{e}_1, \hat{e}_2, \hat{e}_3\end{aligned}$$

We can then define a_{ij} s.t. $a_{1j} = \alpha_j, a_{2j} = \beta_j, a_{3j} = \gamma_j$. We can therefore write

$$\vec{v} = \sum_j x_j \hat{e}_j$$

to simplify notation further, repeated indexes are thus an implicit sum.

$$\vec{v} = v_j \hat{e}_j$$

$$\vec{v} = v'_j \hat{e}'_j$$

Changing indexes now we will have that

$$v'_i = a_{ij}v_j$$

Note that a_{ij} is symmetric and v_j is a column vector which we can write as

$$v' = \begin{pmatrix} v'_x \\ v'_y \\ v'_z \end{pmatrix}$$

Thus we have the matrix equation

$$v' = Av$$

We demand that $\hat{e}'_i \cdot \hat{e}'_j = 0, i \neq j = \delta_{ij}$ where θ_{ij} is actually a tensor and can be represented as the identity matrix. We now have that

$$\hat{e}'_i \cdot \hat{e}'_k = (a_{ij}\hat{e}_j) \cdot (a_{kl}\hat{e}_l) = a_{ij}a_{kl} = \delta_{ik}$$

Now recall that

$$AB = A_{ij}B_{jk}$$

Notice the index difference where we instead have a_{kj} as a transpose. Thus we can show that

$$AA^T = 1$$

$$\exists^{-1} \text{ inverse } A^{-1}(\theta)A(\theta) = 1$$

Thus we must have that

$$A^{2D} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

and we must have that

$$A^T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Note that the product between the two is I. Thus the inverse transformation is simply by a change in the sign of the angle. What is wrong.

Lecture 13

Consider $\hat{n}$ which is towards the rotation axis and the terms $\hat{n} \times rht, \dot{n} \times (\hat{n} \times \hat{r})$.. We have that

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$$\vec{r} = (\vec{r} \cdot \hat{n})\hat{n} + \hat{n} \times \vec{r} + (\vec{r} - (\vec{r} \cdot \hat{n})\hat{n})$$

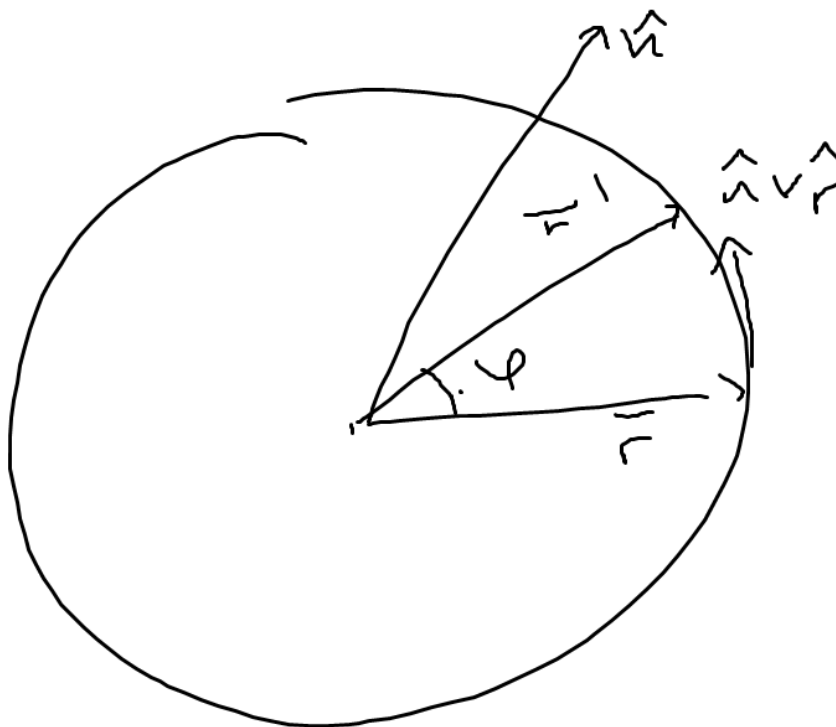


Figure 5.1

$$\vec{r}' = \vec{r} + \hat{n} \times \vec{r} \sin \phi + (\vec{r} - (\vec{r} \cdot \hat{n})\hat{n}) \cos \phi$$

5.1 Infinitesimal Rotations

We can reduce A to an infinitesimal where

$$A \rightarrow A = 1 + \epsilon$$

where ϵ is a rotation matrix to the first element. Recall that we have that

$$\vec{v}' = A\vec{v} = (1 + \epsilon)\vec{v}$$

$$\vec{v}' - \vec{v} = \epsilon\vec{v}$$

Note that for $(1 + \epsilon_1)(1 + \epsilon_2) = 1 + (\epsilon_1 + \epsilon_2) + \mathcal{O}(n^2)$. Notice that

$$A^{-1}A = 1 \Rightarrow A^{-1}1 - \epsilon$$

for infinitesimal transformations since $(1 - \epsilon)(1 + \epsilon) = 1$. We also know that $A^{-1} = A^T$ therefore we have that $A^T = 1 - \epsilon$ which does nothing to the identity matrix. Therefore we have that

$$\epsilon^T = -\epsilon$$

This implies that ϵ is antisymmetric. We must have that

$$\epsilon = \begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$$

however we won't be using this sign convention for reasons we are going to explore. From last lecture we had that for rotation around the z -axis

$$A_z(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Now consider $\theta \ll 1$ we have that the matrix becomes simply

$$\begin{pmatrix} 1 & \delta\theta & \\ -\delta\theta & 1 & \\ & & 1 \end{pmatrix}$$

If we write this as $A = 1 + \epsilon$ and we have that

$$\epsilon_z(\delta\theta) = \delta\theta \begin{pmatrix} & 1 & \\ -1 & & \end{pmatrix}$$

Consider rotations around the y axis we have that

$$A_y(\phi) = \begin{pmatrix} \cos \phi & & -\sin \phi \\ & 1 & \\ \sin \phi & & \cos \phi \end{pmatrix}$$

We will have a generator term where

$$\epsilon_y(\delta\phi) = \delta\phi \begin{pmatrix} & -1 & \\ & & \\ 1 & & \end{pmatrix}$$

We have that

$$\epsilon_x(\delta\alpha) = \delta\alpha \begin{pmatrix} & & \\ & 1 & \\ -1 & & \end{pmatrix}$$

We have that

$$\widetilde{M}_1 = \begin{pmatrix} & \\ & 1 \\ -1 & \end{pmatrix}$$

and the same for $\widetilde{M}_2, \widetilde{M}_3$. This tells us some interesting properties that extend rigid body motion. Consider the following calculation of

$$\widetilde{M}_1\widetilde{M}_2 - \widetilde{M}_2\widetilde{M}_1 = \begin{pmatrix} & \\ 1 & \end{pmatrix} - \begin{pmatrix} 1 & \\ & \end{pmatrix} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix}$$

Remark. This is the same as $\widetilde{M}_3$ up to a minus sign. Note that we are doing passive transformations which implies that the commutator $[\widetilde{M}_1\widetilde{M}_2] = -\widetilde{M}_3$. This is the structure of commutation relations for quantum mechanics! In quantum mechanics, we have active rotations instead but they still have the same form.

5.2 Notation Changes

Let us have the following changes:

$$\delta\theta \rightarrow d\Omega_3 \quad z$$

$$\delta\phi \rightarrow \delta\Omega_2 \quad y$$

$$\delta; a \rightarrow \delta\Omega_1 \quad x$$

We can now write

$$(1 + \epsilon_x)(1 + \epsilon_y)(1 + \epsilon_z) = 1 + \epsilon_x + \epsilon_y + \epsilon_z$$

which is one matrix with all infinitesimal terms. We can write that

$$\epsilon = \begin{pmatrix} & d\Omega_3 & -d\Omega_2 \\ -d\Omega_3 & & d\Omega_1 \\ d\Omega_2 & -d\Omega_1 & \end{pmatrix}$$

We can write the following that

$$d\vec{v} = \epsilon \vec{v}$$

$$dv_1 = d\Omega_3 v_2 - d\Omega_2 v_3$$

$$dv_2 = -d\Omega_3 v_1 + d\Omega_1 v_3$$

Rewrite this we have that

$$d\vec{v} = \vec{v} \times d\vec{\Omega}$$

Now the argument is made that vectors cannot be commuting. However, for infinitesimal transformations do commute. Note that what we get is for passive rotations. For active rotations the result is different. Here in our notation, we switch to active rotations. Therefore we instead have

$$d\vec{v} = -\vec{v} \times d\vec{\Omega}$$

Consider now we have $\frac{d}{dt}$ applied to be

$$\frac{d\vec{v}}{dt} = -\vec{\omega} \times \vec{v}$$

Consider some amorphous object about a fixed point such that it rotates along an axis through that fixed point.

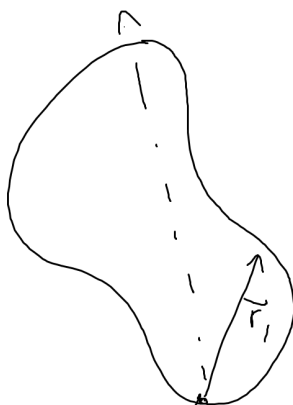


Figure 5.2

Notice that the velocity of every component will be of the form

$$\vec{v}_i = \vec{\omega} \times \vec{r}_i$$

We have

$$\begin{aligned} \vec{L} &= \sum_i m_i \vec{r}_i \times \vec{v}_i \\ &= m_i \vec{r}_i \times (\vec{\omega} \times \vec{r}_i) \\ \vec{L} &= m_i (\vec{r}_i^2 \vec{\omega} - \vec{r}_i (\vec{\omega} \cdot \vec{r}_i)) \\ L_x &= m_i (r_i^2 \vec{\omega}_x - x_i (\omega_x x_i \hat{x} + \omega_y y_i \hat{y} + \omega_z z_i \hat{z})) \\ &= \omega_x m_i (r_i^2 - x_i^2) - m_i x_i y_i \omega_y - m_i x_i z_i \omega_z \end{aligned}$$

Lecture 14

Lecture 15

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Lecture 16

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Lecture 17

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